Extra Practice 1: Solving Linear–Quadratic Systems

A. Skills

1. Solve each system algebraically.

(a) $y = x^2 + 4x - 1,\ \ y = 2x + 7$

(b) $y = -x^2 + 6,\ \ y = x$

(c) $y = 2x^2 + 3x - 1,\ \ y = 7x - 3$

(d) $x - y = 2,\ \ y = x^2 - 5x + 6$

2. Solve by elimination: $y + x^2 = 4$ and $y - 3x = 0$.

3. Verify that $(-1, 4)$ and $(2, -2)$ are the solutions of the system $y = x^2 - 3x$ and $y = -2x + 2$.

4. Without solving, determine the number of solutions of each system.

(a) $y = x^2 + 5,\ \ y = 2x + 4$

(b) $y = -x^2 + 2x,\ \ y = 3$

(c) $y = 4x^2 - x,\ \ y = 3x + 1$

B. Applying

5. Solve $y = x^2 - 2x - 1$ and $y = x + 2$. Give exact answers, then round to the nearest hundredth.

6. Use a graphing calculator to solve $y = -0.5x^2 + 3x + 2$ and $y = x - 1$, to the nearest tenth. State a suitable window.

7. Determine k so that the line $y = -2x + k$ is tangent to the parabola $y = -x^2 + 4x$, and find the point of tangency.

8. Determine the exact points of intersection of the circle $x^2 + y^2 = 40$ and the line $y = 3x$.

9. Which line is tangent to the parabola $y = x^2 + 1$?

A. $y = 2x$ **B.** $y = 2x + 1$ **C.** $y = x$ **D.** $y = 2x - 1$

10. Numerical Response The system $y = x^2 - 6x + 10$ and $y = 2x - c$ has exactly one solution. The value of c is _____.

C. Stretch

11. Determine the values of b for which the line $y = x + 2$ is tangent to the parabola $y = x^2 + bx + 3$. Give the point of tangency for each.

12. Two lines through the point $(0, -4)$ are tangent to the parabola $y = x^2 + 2x$. Determine their equations and the points of tangency.

13. The system $y = x^2 + 2x + k$ and $y = 8x - 1$ has two solutions, and the x-coordinate of one is twice the x-coordinate of the other. Determine k and solve the system.

Extra Practice 2: Solving Quadratic–Quadratic Systems

A. Skills

1. Solve each system algebraically.

(a) $y = x^2 + x - 3,\ \ y = -x^2 + 3x + 1$

(b) $y = 2x^2 - 4x + 1,\ \ y = x^2 - x + 5$

(c) $y = x^2 + 6x,\ \ y = x^2 - 2x + 16$

(d) $y = 3x^2 - 2x - 1,\ \ y = x^2 + x + 1$

2. Without solving, state the number of solutions of each system.

(a) $y = x^2 + 4,\ \ y = -x^2 + 4x + 2$

(b) $y = -x^2 + 1,\ \ y = x^2 + 2x + 3$

(c) $y = 5x^2 - 3x + 2,\ \ y = 5x^2 - 3x + 7$

(d) $y = -2(x + 1)^2 + 4,\ \ y = -2x^2 - 4x + 2$

3. Verify that $(3, 4)$ is a solution of the system $y = x^2 - 4x + 7$ and $y = -x^2 + 6x - 5$, then find the other solution.

B. Applying

4. Solve $y = x^2 - 1$ and $y = -x^2 + 4x + 3$ exactly, then give the solutions to the nearest tenth (check with the calculator's intersect).

5. Solve: $y - 2x^2 = x - 3$ and $y + x^2 = 4x + 3$.

6. Solve $y = \frac{1}{4}x^2 + x$ and $y = -\frac{1}{2}x^2 + 4x - 3$, and describe how the graphs meet.

7. Determine k so that the system $y = x^2 - 2x + k$ and $y = -x^2 + 6x$ has exactly one solution, and find it.

8. The parabola $y = x^2$ passes through $(1, 1)$ and $(2, 4)$. Write the equation of a different parabola, opening down, that meets $y = x^2$ at exactly these two points. Verify.

9. Which system has no solution?

A. $y = x^2 + 3$ and $y = x^2 - 3$

B. $y = x^2 + 3$ and $y = -x^2 + 5$

C. $y = x^2 + 3$ and $y = 2x^2 - 1$

D. $y = x^2 + 3$ and $y = (x - 1)^2$

10. Numerical Response The greater y-coordinate of the solutions of the system $y = 2x^2 - 5x + 4$ and $y = x^2 + x - 4$ is _____.

C. Stretch

11. Determine all values of a for which the system $y = ax^2 + 2x$ and $y = x^2 - 2x - 1$ has exactly one solution, and give the solution for each.

12. Determine p and q so that the system $y = 3(x - p)^2 + q$ and $y = 3x^2 - 12x + 7$ has infinitely many solutions.

Extra Practice 3: Problems Involving Systems

A. Skills

1. The sum of two numbers is 9, and the larger number is 3 more than the square of the smaller number. Write a system and determine both possible pairs.

2. A rectangular tablecloth has an area of 84 dm^2 and a perimeter of 38 dm. Determine its dimensions.

3. Two balls are thrown upward at the same moment. Their heights in metres after t seconds are $h = -5t^2 + 18t + 8$ and $h = -5t^2 + 22t$. When are they at the same height, and what is that height?

4. Determine k so that the line $y = -2x + k$ is tangent to the parabola $y = x^2 + 4x + 1$, and give the point of tangency.

B. Applying

5. A stone is tossed from the foot of a hill. Its path is $y = -0.1x^2 + 3x$ and the hillside is the line $y = x$ (metres). Where does the stone land, and how far up the hillside is that, to the nearest tenth of a metre?

6. A lawn sprinkler at the origin waters every point within 6 m, the region inside $x^2 + y^2 = 36$. A straight garden path runs along the line $x + y = 6$. Determine where the path enters and leaves the watered region, and the length of path that gets wet, to the nearest tenth of a metre.

7. A bakery's daily revenue and cost for x hundred loaves are $R = -3x^2 + 48x$ and $C = x^2 + 8x + 64$ (in hundreds of dollars).

(a) Determine the break-even quantities.

(b) Determine the maximum profit.

8. A ball's height is $h = -5(t - p)^2 + q$ metres after t seconds. After 2 s it is 55 m high and after 5 s it is 40 m high. Determine p and q, the starting height, and the time the ball lands, to the nearest hundredth of a second.

9. The line $y = x + c$ does **not** intersect the parabola $y = x^2 + 3x + 2$ when

A. $c < 1$ **B.** $c > 1$ **C.** $c < -1$ **D.** $c = 1$

10. Numerical Response Two positive integers differ by 2 and the sum of their squares is 244. The product of the two integers is _____.

C. Stretch

11. The system $y = x^2 - 8x + 20$ and $y = 4x + k$ has two solutions, and the x-coordinate of one is twice the x-coordinate of the other. Determine k and both solutions.

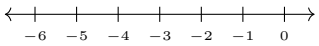
12. Two water jets in a fountain follow the paths $y = -0.5x^2 + 4x$ and $y = -x^2 + 10x - 16$ (metres). Determine where the jets cross, and explain what the point with $y = 0$ means.

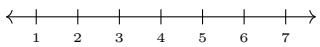
13. Solve the system $x^2 + y^2 = 25$ and $y = x^2 - 5$ algebraically. (Hint: substitute $x^2 = y + 5$.)

Extra Practice 4: Linear Inequalities in Two Variables

A. Skills

1. Solve, graph on the number line and write in set notation.
(a) $7 - 4x \geq 19$ (b) $2(x + 5) < 6x - 6$

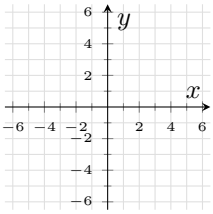




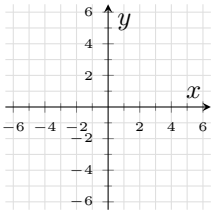
2. Which of the points $(0, 0)$, $(2, 2)$, $(5, -2)$ and $(-1, 5)$ are solutions of $y > -x + 3$?

3. Write each inequality in the form $y < mx + b$, $y \leq mx + b$, $y > mx + b$ or $y \geq mx + b$. State whether the boundary is solid or dashed and whether the region is above or below it. (a) $5x - y < 3$ (b) $-4x - 2y \geq 8$

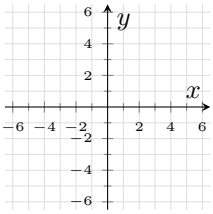
4. Graph each inequality.
(a) $y \geq -2x + 4$



(b) $x - 3y > 6$



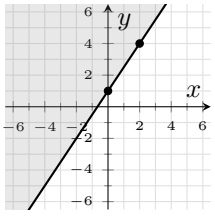
(c) $2y + 5 \geq 0$



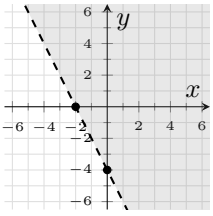
B. Applying

5. Write the inequality whose solution region is shown.

(a)



(b)

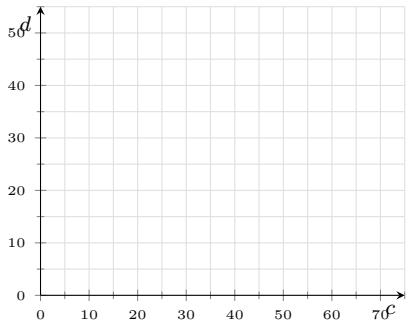


6. The boundary of a solution region is a solid line through $(2, 5)$ and $(-4, -1)$, and the region lies below the line. Write the inequality.

7. A team raises money by washing cars for \$8 and dogs for \$12. Its goal is at least \$480. Let c be the number of cars and d the number of dogs.

(a) Write an inequality, in simplest form, with restrictions.

(b) Graph the solution region.



(c) If the team washes 30 dogs, how many cars must it wash?

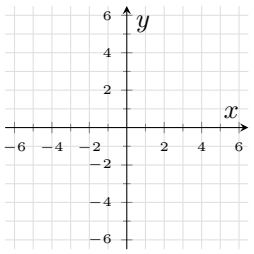
8. Which point is in the solution region of $2x - 5y > 10$?

A. $(0, 0)$ **B.** $(5, 0)$ **C.** $(0, -2)$ **D.** $(4, -1)$

9. Numerical Response The point $(-5, 6)$ lies on the boundary line of the inequality $3x + ky \geq 15$. The value of k is _____.

C. Stretch

10. Graph the region that satisfies $y \geq x - 3$, $y \leq -2x + 6$ and $x \geq 0$. Determine the corners of the region and its area.



11. A woodworker builds chairs (3 h each) and tables (5 h each). She has at most 30 hours this week and must build at least 2 tables. Let c be the number of chairs and t the number of tables.

(a) Write the system of inequalities, with restrictions.

(b) List every possible combination (c, t).

(c) She earns \$50 profit per chair and \$70 per table. Which combination gives the greatest profit?

12. Write a system of inequalities whose solution region is the triangle with corners $(0, 0)$, $(4, 0)$ and $(0, 4)$, including its sides.

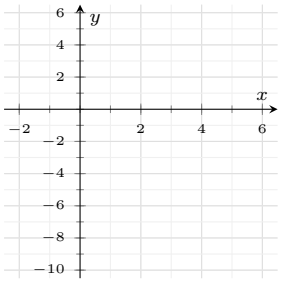
Extra Practice 5: Quadratic Inequalities in Two Variables

A. Skills

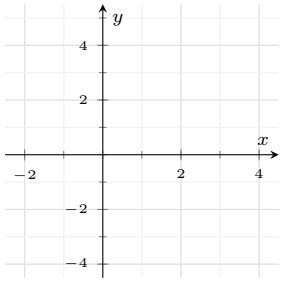
1. Determine whether each point is a solution of $y \leq -x^2 + 4x + 1$.
(a) $(0, 0)$ (b) $(2, 5)$ (c) $(3, 6)$ (d) $(-1, -5)$

2. Write each inequality with y alone on the left. State whether the boundary is solid or dashed and whether the region is above or below the parabola.
(a) $y - 5 < 2x^2$ (b) $3x^2 - y \leq 6$ (c) $-4y > 8x^2 - 4$

3. Graph the solution region of each inequality.
(a) $y \leq x^2 - 4x - 5$

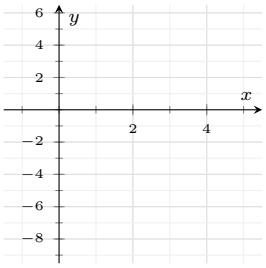


(b) $y > -(x - 1)^2 + 4$

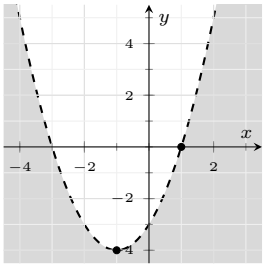


B. Applying

4. Graph the solution region of $y \geq 2x^2 - 8x$. Explain your choice of test point.



5. The boundary of the shaded region has vertex $(-1, -4)$ and passes through $(1, 0)$. Write the inequality in general form, and check it with a point in the region.



6. A solid parabola has x-intercepts -2 and 4 and passes through $(0, -4)$. The solution region is above it.

(a) Write the inequality in general form.

(b) Is $(1, -4)$ a solution?

7. The opening of a tunnel is a parabolic arch 10 m wide at the road and 5 m high in the middle. Put the origin at the centre of the road.

(a) Write the equation of the arch.

(b) Write the inequalities for the points on or under the arch and on or above the road.

(c) Will a load 4 m wide and 4.1 m high, centred, fit through?

(d) How high can a centred load 6 m wide be?

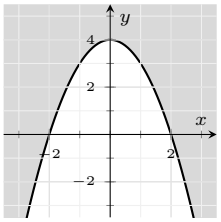
8. The boundary of the shaded region has vertex $(0, 4)$ and x-intercepts ± 2. The inequality is

A. $y \geq -x^2 + 4$

B. $y \leq -x^2 + 4$

C. $y > -x^2 + 4$

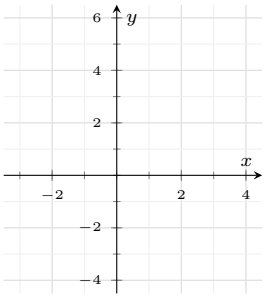
D. $y \geq x^2 - 4$



9. Numerical Response The point $(-2, k)$ is in the solution region of $y < -x^2 + 3x + 15$, and k is an integer. The largest possible value of k is _____.

C. Stretch

10. (a) Shade the region that satisfies both $y \geq x^2 - 4$ and $y \leq x + 2$, and find where the boundaries meet.



(b) How many points with integer coordinates and $x = 1$ are in the region?

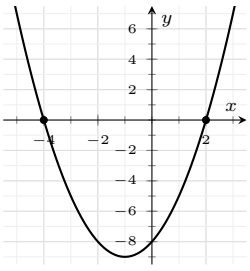
11. For what values of a is the point $(2, 3)$ in the solution region of $y > ax^2 - 1$?

12. A parabola opens down, has x-intercepts 1 and 5, and has vertex $(3, 8)$. Write the inequality whose solution region is the region outside this parabola, boundary included.

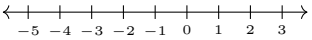
Extra Practice 6: Quadratic Inequalities in One Variable: Roots and Test Points

A. Skills

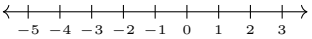
1. The graph of $y = x^2 + 2x - 8$ is shown. Use it to solve each inequality, and show the solution on a number line.



(a) $x^2 + 2x - 8 < 0$



(b) $x^2 + 2x - 8 \geq 0$



2. Solve using roots and test points.

(a) $x^2 - x - 20 > 0$

(b) $x^2 + 8x + 12 \leq 0$

(c) $3x^2 - 12x < 0$

(d) $5x^2 + 13x - 6 \geq 0$

3. Solve each inequality.

(a) $x^2 - 10x + 25 < 0$

(b) $x^2 + x + 3 \geq 0$

(c) $(x + 1)^2 > 0$

B. Applying

4. Rearrange, then solve.

(a) $3x^2 \geq 2 - 5x$

(b) $x(x - 6) < -8$

5. Solve $x^2 + 3x - 5 \leq 0$ (a) with a graphing calculator, to the nearest tenth, and (b) exactly.

6. Solve $(x+5)(x-3)<0$ using roots and test points.

7. For what values of k does $x^2+kx+16=0$ have no real roots?

8. The solution of $-x^2+6x-5\geq 0$ is

A. $x\leq 1$ or $x\geq 5$ **B.** $1\leq x\leq 5$ **C.** $-5\leq x\leq -1$ **D.** $1<x<5$

9. Numerical Response The number of integers that satisfy $x^2-2x-24<0$ is _____.

C. Stretch

10. Solve $2(x-1)^2\geq (x+1)^2$ exactly.

11. Write a quadratic inequality, in the form $ax^2+bx+c>0$ with integer coefficients, whose solution is

$$\left\{x \mid x<-\tfrac{1}{2} \text{ or } x>3, x\in\mathbb{R}\right\}.$$

12. A square has side x cm. A rectangle has length $(2x-3)$ cm and width $(x+2)$ cm. For what values of x is the area of the rectangle greater than the area of the square?

Extra Practice 7: Quadratic Inequalities in One Variable: Sign Analysis

Write each solution in set notation.

A. Skills

1. Solve $(x-5)(x+1) \geq 0$ by case analysis.

2. Solve with a sign chart.
(a) $x^2 - 8x + 12 < 0$

(b) $x^2 + 5x \geq 0$

(c) $4x^2 - 25 > 0$

(d) $3x^2 - 7x - 6 \leq 0$

3. Rearrange so one side is 0, then solve.
(a) $2x^2 > 9x + 5$

(b) $(x+2)(x-4) \leq 7$

(c) $3x^2 - x - 4 \geq x^2 + 2x + 5$

4. Solve.
(a) $18 + 3x - x^2 > 0$

(b) $-6x^2 + 13x - 6 \geq 0$

5. Solve.
(a) $x^2 + 14x + 49 > 0$ (b) $-x^2 + 6x - 10 < 0$ (c) $25x^2 - 20x + 4 \leq 0$

B. Applying

6. Write a quadratic inequality with integer coefficients and a positive leading coefficient for each solution.
(a) $\{x \mid -7 \leq x \leq 1, x \in \mathbb{R}\}$ (b) $\{x \mid x < \frac{2}{3} \text{ or } x > 5, x \in \mathbb{R}\}$

7. A diver jumps up from a 10 m platform. Her height above the water after t seconds is $h = -5t^2 + 5t + 10$ metres.
(a) When does she enter the water?

(b) When is she above the platform?

(c) When is she less than 5 m above the water? Round to the nearest hundredth.

8. A farmer uses 30 m of fencing for three sides of a rectangular pen; a barn wall forms the fourth side. Each side touching the wall is x m long. For what values of x is the area at least 100 m^2?

9. For what values of k does $x^2 + (k+1)x + 4 = 0$ have two distinct real roots?

10. Which inequality has the solution $\{x \mid -1 < x < 6, x \in \mathbb{R}\}$?
A. $x^2 - 5x - 6 > 0$ **B.** $x^2 - 5x - 6 < 0$ **C.** $x^2 + 5x - 6 < 0$ **D.** $-x^2 + 5x + 6 < 0$

11. Numerical Response The greatest integer that satisfies $2x^2 - 11x - 21 < 0$ is _____.

C. Stretch

12. Solve $(x+3)(x-1)(x-5) < 0$.

13. Solve $x^2 - 6x + 4 \geq 0$. Give exact endpoints.

14. For what values of c is $x^2 + 6x + c > 0$ true for every real number x? Explain.

Extra Practice 8: Unit 8 Review — Practice Test

Calculators are allowed. Give exact answers unless a question asks you to round. Write one-variable solutions in set notation.

A. Multiple Choice

1. The number of solutions of the system $y = -x^2 + 4x - 1$ and $y = x + 3$ is

A. 0 **B.** 1 **C.** 2 **D.** infinitely many

2. The solution of the system $y = x^2 - 1$ and $y = x + 1$ is

A. $(2, 3)$ and $(-1, 0)$

B. $(-2, -1)$ and $(1, 2)$

C. $(2, 3)$ only

D. $(-1, 0)$ only

3. The system $y = x^2 + 3x - 2$ and $y = x^2 - x + 6$ has

A. no solution **B.** the solution $(2, 8)$ **C.** the solution $(-2, -4)$ **D.** infinitely many solutions

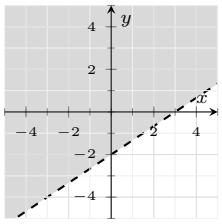
4. The line $y = 4x + k$ is tangent to the parabola $y = x^2$ when k is

A. 4 **B.** -4 **C.** 16 **D.** -16

5. The point that is in the solution region of $3x + 2y > 6$ is

A. $(0, 3)$ **B.** $(1, 1)$ **C.** $(2, 1)$ **D.** $(-1, 4)$

6. The dashed boundary passes through $(0, -2)$ and $(3, 0)$. The shaded region is the solution of



A. $2x - 3y < 6$ **B.** $2x - 3y > 6$

C. $2x - 3y \leq 6$ **D.** $3x - 2y < 6$

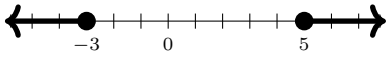
7. Which point **cannot** be used as a test point to graph $y \geq x^2 - 5x$?

A. $(0, 0)$ **B.** $(1, 1)$ **C.** $(0, 2)$ **D.** $(5, 1)$

8. The solution of $x^2 - 2x - 24 < 0$ is

A. $-6 < x < 4$ **B.** $-4 < x < 6$ **C.** $x < -4$ or $x > 6$ **D.** $x < -6$ or $x > 4$

9. The solution shown on the number line is the solution of



A. $(x+3)(x-5) \leq 0$ **B.** $(x-3)(x+5) \geq 0$ **C.** $(x+3)(x-5) \geq 0$ **D.** $(x-3)(x+5) \leq 0$

10. Mia solves $2x^2 + x - 6 \leq 0$ with a sign chart. The values of x where her factor rows change sign are

A. -2 and $\frac{3}{2}$ **B.** 2 and $-\frac{3}{2}$ **C.** -2 and 3 **D.** -6 and 1

B. Numerical Response

11. Numerical Response To the nearest hundredth, the positive x-coordinate of a solution of the system $y = 2x^2 + x - 5$ and $y = x^2 + 4x - 3$ is _____.

12. Numerical Response The line $y = kx + 3$, with $k > 0$, is tangent to the parabola $y = x^2 + 2x + 7$. The value of k is _____.

13. Numerical Response A craft seller's profit, in dollars, for n items is $P = -2n^2 + 60n - 352$. The least number of items she must sell to make a profit ($P > 0$) is _____.

14. Numerical Response The number of integers that satisfy $x^2 + 3x - 28 \leq 0$ is _____.

C. Written Response

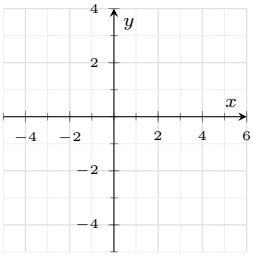
15. Solve each system algebraically and verify.

(a) $y = x^2 - 2x - 5,\ \ y = 3x + 1$

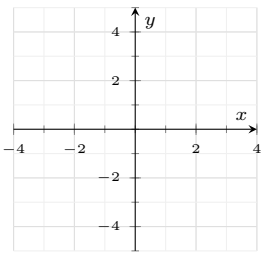
(b) $y = 2x^2 - 5x + 1,\ \ y = -x^2 + x + 10$

16. Graph each inequality.

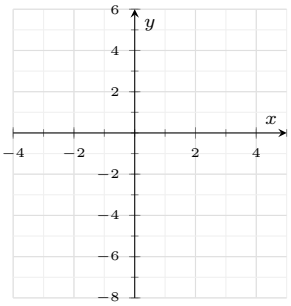
(a) $x - 2y \geq 4$



(b) $y \leq -x^2 + 4$



17. (a) Describe how to graph an inequality of the form $y \geq ax^2 + bx + c$ without technology.



(b) Graph $y \geq x^2 - x - 6$ on the grid.

(c) Use your graph to solve $x^2 - x - 6 < 0$. Explain.

18. At the same moment, a ball is thrown upward from the ground and another from a 20 m roof. Their heights in metres after t seconds are $h = -5t^2 + 30t$ and $h = -5t^2 + 20t + 20$.

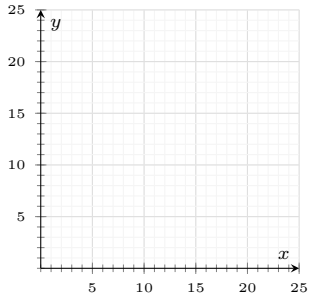
(a) When are the balls at the same height, and what is that height?

(b) During what time interval is the first ball higher than 40 m?

19. Lena earns \$20/h tutoring and \$15/h lifeguarding. She wants to earn at least \$300 a week. Let x be her tutoring hours and y her lifeguarding hours.

(a) Write an inequality, with restrictions.

(b) Graph the solution region.



(c) Give two combinations of hours that meet her goal, one of them on the boundary.